Santiago Sosa

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Education

Drexel University | Philadelphia, PA *September 2021 – Present*
Master of Science in Electrical Engineering *Anticipated Graduation: June 2026*
Bachelor of Science in Electrical Engineering *Anticipated Graduation: June 2026*
Minor: Computer Science

- BS/MS accelerated degree program
- Penonni Honors College
- SuperNova Undergraduate Research Fellow

Skills

Programming: Python (Tensorflow, PyTorch, Pandas), C/C++, Embedded C, MATLAB, VHDL, Arduino, R, ROS
Platforms: MacOS, Unix, Windows
Hardware: FIBSEM, SEM-EDS, CCAs, Oscilloscope, Logic analyzer, Spectrometer, Lasers, Spectrum Analyzer, Microcontrollers
Software: Altium, LTSpice, Microsoft (Word, Excel, PowerPoint), AutoCAD Fusion 360, ImageJ, Tuscen, OceanView
Communication: Design proposals, Technical reports, Presentations, Client meetings, Design Reviews
Languages: English (Native), Spanish (Native)
Clearance: DoD Secret

Experience

Lockheed Martin | Cherry Hill, NJ *September 2023 – March 2024*
Research Engineering Co-op Sr / Advanced Technology Labs (Spectral Systems Laboratory)

- Researching state-of-the-art defense technology at Advanced Technology Labs, Spectral Systems Laboratory
- Developing novel **convolutional neural network** systems to enhance **object tracking**
- Innovating RF communication solution for **MIMO** system
- Researching watermarking techniques and **waveform encryption**, with GAN encoders

Lockheed Martin | Moorestown, NJ *September 2022 – March 2023*
Power System Engineering Co-op / Power Subsystems (Integrated Warfare Systems and Sensors – RMS)

- Collaborated as member of **tiger team** to optimize Space Fence radar sensitivity, and create alternate circuit capabilities
- Performed **testing and analysis** of transmit hardware to enhance Space Fence radar for **low earth orbit object tracking**
- Enhanced ASEV (Aegis System Equipped Vessels) radar processor power supply and input filter for calibration
- Traveled to Space Fence radar site on Kwajalein Atoll to perform testing on transmit assembly and power hardware

Research

Drexel University Electrical and Computer Engineering | Philadelphia, PA *April 2022 – Present*
Researcher / NanoPhotonics+ Laboratory (Advisor: Dr. Adam K. Fontecchio, PhD)

- Developing innovative **hyperspectral imaging** system able to capture images in color and infrared spectrum
- Programming autonomous vehicle **decision making** using **color filtering** and **image segmentation**
- Creating H-PDLCs (Holographic Polymer Dispersed Liquid Crystals) as switchable Bragg grating structures to enhance **chemical trace detection** system

Consulting

Fontecchio Consulting | Philadelphia, PA *June 2023 – Present*
Consultant

- Conducting comprehensive analysis of modern nanotechnology in phones for patent litigation expert testimony report

Relevant Coursework

- Probability and Random Variables (ECES 521)
- Medical Robotics (ECES 561)
- Applied Machine Learning Engineering (ECE 612)
- Decision Making for Robotics (ECE 608)

Presentations

STAR Scholar Research Program *September 2022*

- "Hyperspectral Imaging System Through Holographic Polymer Dispersed Liquid Crystals (H-PDLCs) Switchable Filters" Santiago Sosa, Marquise Pullen, Tyler Hein, and Adam K. Fontecchio, STAR Poster Session, September 2022

Activities

SuperNova Undergraduate Research Fellows Program *September 2023 - Present*

- Challenges undergraduate students to build upon their STAR or undergraduate research experience